

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration,Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks:50

Annexure A SENARIO BASED QUESTIONS

Criteria	Understanding of Scenario (2)	Identification of Key Issues (2)	Application of Concepts (2.5)	Decision Making (2)	Clarity of Response (1.5)	Total(10)
Weightage	20%	20%	25%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I, G Madhulatha(192311295), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

G Madhulatha

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Scenario based Questions

Q1. Data Preprocessing in Retail Analytics (L3, L4)

A **retail analytics company** is preparing customer purchase data for building a recommendation system. The dataset contains missing values, inconsistent categorical formats, and extreme values due to data collected from multiple stores and manual entries.

Sample Dataset

Customer_ID	Age	Gender	Monthly_Spend (₹K)	Purchase_Frequency	Loyalty_Level
C1	23	M	20	5	Low
C2	NaN	Male	35	8	Medium
C3	31	F	NaN	10	High
C4	105	Female	500	50	High
C5	27	male	25	NaN	Low
C6	40	FEMALE	45	12	Medium
C7	36	M	38	9	Low
C8	42	Female	300	45	High

Tasks

A. Handling Missing Values

- Identify missing values in **Age, Monthly_Spend, and Purchase_Frequency**
 - Apply at least **two imputation techniques** (e.g., Median Imputation, k-NN Imputation)
 - Compute the imputed values and justify the selected method
-

B. Outlier Detection and Treatment

- Detect outliers in **Age and Monthly_Spend** using the **Z-score method**
 - Identify records containing extreme values (e.g., Age = 105, Spend = 500)
 - Apply suitable treatment methods (**capping / transformation / removal**) and justify
-

C. Categorical Data Standardization

- Identify inconsistencies in **Gender** (M, Male, male, FEMALE, etc.)
- Convert all entries into a standardized format (**Male / Female**)

Q2. Covariance & Data Reduction in E-commerce Analytics (L4, L5)

An **e-commerce platform** is building a predictive model to classify high-value customers. The dataset includes several correlated financial and behavioral attributes, leading to redundancy and increased computational cost.

Sample Dataset

Cust_ID	Age	Income (₹K)	Purchase_Value (₹K)	Credit_Score	Orders	Avg_Balance (₹K)	EMI (₹)	Customer_Class
U1	24	40	100	650	15	12	5000	Low
U2	34	60	150	700	20	18	7000	Medium
U3	44	85	210	720	28	25	9000	Medium
U4	52	120	320	800	35	40	15000	High
U5	29	45	110	680	18	14	5500	Low
U6	39	75	180	710	24	22	8500	Medium
U7	56	150	360	820	40	50	18000	High
U8	37	70	160	705	22	20	8000	Medium

Tasks

A. Covariance Analysis

- Compute covariance between **Income and Purchase_Value** (show steps or partial calculation)
 - Interpret whether the relationship is **positive or negative**
 - Explain how covariance helps in identifying feature relationships
-

B. Feature Selection

- Identify redundant attributes using **correlation or domain knowledge**
 - Select an optimal subset of features for modeling
 - Justify your selection
-

C. Data Transformation

- Standardize the dataset (excluding **Cust_ID** and **Customer_Class**)
- Explain why normalization/standardization is required

Q3. Data Transformation & Discretization in Education Analytics (*L3, L4, L5*)

An **education analytics team** is developing a model to predict student performance. Continuous variables such as study hours and test scores need to be discretized to improve interpretability and decision-making.

Sample Dataset

Student_ID	Study_Hours	Test_Score	Attendance (%)	Performance
S1	2	40	60	Poor
S2	3	50	65	Average
S3	4	55	70	Average
S4	5	65	75	Good
S5	6	75	80	Good
S6	7	85	85	Good
S7	8	90	88	Excellent
S8	9	95	90	Excellent
S9	10	98	92	Excellent
S10	11	100	95	Excellent

Tasks

A. Equal-Width Binning

- Apply **3 bins** for Study_Hours and Test_Score
 - Calculate bin width and define intervals
-

B. Equal-Frequency Binning

- Divide data into **3 bins with equal records**
 - Assign values accordingly
-

C. Concept Hierarchy Construction

- Study_Hours → Low / Medium / High

- Test_Score → Poor / Average / Excellent

4. Use the two methods below to normalize the following group of data:

200, 300, 400, 600, 1000

(a) min-max normalization by setting min = 0 and max = 1

(b) z-score normalization

5. Data Transformation in Online Shopping Behavior Analysis (L3, L4, L5)

An **e-commerce company** is analyzing customer engagement by measuring the **time spent (in seconds)** on product pages. The duration varies significantly—from a few seconds to more than an hour—resulting in highly skewed data.

To improve the effectiveness of **customer segmentation using clustering techniques**, the dataset must be properly transformed and normalized.

Sample Dataset: Visit Duration

Customer_ID Visit_Duration (seconds)

C1	15
C2	30
C3	45
C4	60
C5	120
C6	300
C7	600
C8	900
C9	1800
C10	3600

Tasks

A. Equal-Frequency Binning

- Divide the dataset into **3 bins** with approximately equal number of records
 - Assign each value to its respective bin
 - Analyze how effectively this method groups short and long visit durations
-

B. Z-Score Normalization

- i. Compute the **mean and standard deviation** of visit duration
- ii. Apply Z-score normalization:

$$Z = \frac{X - \mu}{\sigma}$$

- iii. Discuss how this method handles **extreme values (e.g., 1800, 3600 seconds)**
-

C. Min-Max Normalization

- i. Apply Min-Max normalization using range **[0,1]**:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

- ii. Transform all values accordingly
- iii. Compare results with Z-score normalization in terms of **scaling and sensitivity to outliers**

DATA PREPROCESSING, COVARIANCE, DATA REDUCTION AND TRANSFORMATION

Q1. DATA PREPROCESSING IN RETAIL ANALYTICS (L3, L4)

A. HANDLING MISSING VALUES

i. Identification of Missing Values

The given dataset contains the following missing values:

Age:

C2 has a missing Age value.

Monthly Spend:

C3 has a missing Monthly Spend value.

Purchase Frequency:

C5 has a missing Purchase Frequency value.

Therefore, there are 3 missing values in total.

ii. Imputation Techniques

TECHNIQUE 1: MEDIAN IMPUTATION

Median imputation is suitable because the dataset contains extreme values such as Age = 105 and Monthly Spend = 500.

1. Age Imputation

Available Age values:

23, 31, 105, 27, 40, 36, 42

Sorted values:

23, 27, 31, 36, 40, 42, 105

Median = 36

Therefore,

C2 Age = 36

2. Monthly Spend Imputation

Available Monthly Spend values:

20, 35, 500, 25, 45, 38, 300

Sorted values:

20, 25, 35, 38, 45, 300, 500

Median = 38

Therefore,

C3 Monthly Spend = ₹38K

3. Purchase Frequency Imputation

Available Purchase Frequency values:

5, 8, 10, 50, 12, 9, 45

Sorted values:

5, 8, 9, 10, 12, 45, 50

Median = 10

Therefore,

C5 Purchase Frequency = 10

TECHNIQUE 2: MEAN IMPUTATION

Mean Age:

Mean = $(23 + 31 + 105 + 27 + 40 + 36 + 42) / 7$

Mean = $304 / 7$

Mean = 43.43

Mean Monthly Spend:

Mean = $(20 + 35 + 500 + 25 + 45 + 38 + 300) / 7$

Mean = $963 / 7$

Mean = 137.57

Mean Purchase Frequency:

$$\text{Mean} = (5 + 8 + 10 + 50 + 12 + 9 + 45) / 7$$

$$\text{Mean} = 139 / 7$$

$$\text{Mean} = 19.86$$

Comparison of Imputation Methods:

Age:

$$\text{Median} = 36$$

$$\text{Mean} = 43.43$$

Monthly Spend:

$$\text{Median} = 38$$

$$\text{Mean} = 137.57$$

Purchase Frequency:

$$\text{Median} = 10$$

$$\text{Mean} = 19.86$$

iii. Selected Method and Justification

Median imputation is selected.

Reasons:

1. The dataset contains extreme values such as Age = 105 and Monthly Spend = ₹500K.
2. Mean values are strongly affected by extreme values.
3. Median is more robust to outliers.

4. Therefore, median imputation provides more reliable representative values.

FINAL IMPUTED DATASET

Customer ID	Age	Gender	Monthly Spend	Purchase Frequency	Loyalty Level
C1	23	Male	20	5	Low
C2	36	Male	35	8	Medium
C3	31	Female	38	10	High
C4	105	Female	500	50	High
C5	27	Male	25	10	Low
C6	40	Female	45	12	Medium
C7	36	Male	38	9	Low
C8	42	Female	300	45	High

B. OUTLIER DETECTION AND TREATMENT

i. Z-Score Method

Formula:

$$Z = (X - \text{Mean}) / \text{Standard Deviation}$$

AGE

Mean Age = 43.43

Standard Deviation = 25.92

For Age = 105:

$$Z = (105 - 43.43) / 25.92$$

$$Z = 61.57 / 25.92$$

$$Z = 2.38$$

Therefore, Age = 105 is identified as an extreme value.

MONTHLY SPEND

$$\text{Mean} = 137.57$$

$$\text{Standard Deviation} = 174.54$$

For Monthly Spend = 500:

$$Z = (500 - 137.57) / 174.54$$

$$Z = 362.43 / 174.54$$

$$Z = 2.08$$

Therefore, Monthly Spend = ₹500K is identified as an extreme value.

ii. Records Containing Extreme Values

C4 contains:

$$\text{Age} = 105$$

Monthly Spend = ₹500K

C8 also contains a high Monthly Spend value of ₹300K.

iii. Outlier Treatment

A suitable method is capping or transformation instead of removing the records.

Reasons:

1. Extreme values may contain useful customer information.
2. Removing C4 completely may result in loss of valuable data.
3. Capping limits the influence of extreme values.
4. Log transformation can also be applied to Monthly Spend.

Selected treatment:

Capping and/or logarithmic transformation.

C. CATEGORICAL DATA STANDARDIZATION

The Gender column contains:

M

Male

F

Female

male

FEMALE

These values represent only two categories.

Standardization Rules:

M, Male, male $\rightarrow$ Male

F, Female, FEMALE $\rightarrow$ Female

STANDARDIZED GENDER DATA

C1	M	$\rightarrow$ Male
C2	Male	$\rightarrow$ Male
C3	F	$\rightarrow$ Female
C4	Female	$\rightarrow$ Female
C5	male	$\rightarrow$ Male
C6	FEMALE	$\rightarrow$ Female
C7	M	$\rightarrow$ Male
C8	Female	$\rightarrow$ Female

CONCLUSION

Data preprocessing improves data quality by handling missing values, reducing the effect of outliers, and making categorical values consistent. This produces cleaner data for recommendation-system development.

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Q2. COVARIANCE AND DATA REDUCTION IN E-COMMERCE ANALYTICS (L4, L5)

A. COVARIANCE ANALYSIS

We need to calculate covariance between Income and Purchase Value.

Given Data:

Customer	Income (₹K)	Purchase Value (₹K)
U1	40	100
U2	60	150
U3	85	210
U4	120	320
U5	45	110
U6	75	180
U7	150	360
U8	70	160

STEP 1: CALCULATE MEAN

Mean Income:

$$\bar{X} = (40 + 60 + 85 + 120 + 45 + 75 + 150 + 70) / 8$$

$$\bar{X} = 645 / 8$$

$$\bar{X} = 80.625$$

Mean Purchase Value:

$$\bar{Y} = (100 + 150 + 210 + 320 + 110 + 180 + 360 + 160) / 8$$

$$\bar{Y} = 1590 / 8$$

$$\bar{Y} = 198.75$$

STEP 2: CALCULATE COVARIANCE

Formula:

$$\text{Cov}(X,Y) = \Sigma[(X_i - \bar{X})(Y_i - \bar{Y})] / n$$

Partial Calculation:

U1:

$$\begin{aligned} & (40 - 80.625)(100 - 198.75) \\ & = (-40.625)(-98.75) \\ & = 4010.94 \end{aligned}$$

U2:

$$\begin{aligned} & (60 - 80.625)(150 - 198.75) \\ & = (-20.625)(-48.75) \\ & = 1005.47 \end{aligned}$$

U3:

$$\begin{aligned} & (85 - 80.625)(210 - 198.75) \\ & = (4.375)(11.25) \\ & = 49.22 \end{aligned}$$

U4:

$$\begin{aligned}
 &(120 - 80.625)(320 - 198.75) \\
 &= (39.375)(121.25) \\
 &= 4774.22
 \end{aligned}$$

$$\begin{aligned}
 &U5: \\
 &(45 - 80.625)(110 - 198.75) \\
 &= (-35.625)(-88.75) \\
 &= 3160.31
 \end{aligned}$$

$$\begin{aligned}
 &U6: \\
 &(75 - 80.625)(180 - 198.75) \\
 &= (-5.625)(-18.75) \\
 &= 105.47
 \end{aligned}$$

$$\begin{aligned}
 &U7: \\
 &(150 - 80.625)(360 - 198.75) \\
 &= (69.375)(161.25) \\
 &= 11184.84
 \end{aligned}$$

$$\begin{aligned}
 &U8: \\
 &(70 - 80.625)(160 - 198.75) \\
 &= (-10.625)(-38.75) \\
 &= 411.72
 \end{aligned}$$

$$\text{Sum of products} = 24706.25$$

$$\text{Cov}(X,Y) = 24706.25 / 8$$

$$\text{Cov}(X,Y) = 3088.28$$

FINAL ANSWER:

Covariance = 3088.28

ii. Interpretation

Since covariance is positive:

Income and Purchase Value have a POSITIVE relationship.

This means that customers with higher income generally tend to have higher purchase values.

iii. Importance of Covariance

Covariance helps to:

1. Identify whether two attributes move together.
2. Detect positive or negative relationships.
3. Identify potentially redundant features.
4. Support feature-selection decisions.
5. Understand relationships between financial and behavioral variables.

B. FEATURE SELECTION

i. Redundant Attributes

Potentially redundant attributes include:

1. Income
2. Purchase Value

3. Orders
4. Average Balance
5. EMI

Income and Purchase Value are strongly related.

Income and Average Balance also show a strong relationship.

Average Balance and EMI can also contain overlapping financial information.

Purchase Value and Orders are related to customer purchasing behavior.

ii. Selected Features

A suitable feature subset is:

1. Age
2. Income
3. Purchase Value
4. Credit Score
5. Orders

Cust_ID is excluded because it is only an identifier.

Customer_Class is the target variable and should not be used as an input feature.

iii. Justification

The selected features provide information about:

1. Customer demographics.
2. Financial capacity.
3. Purchase behavior.
4. Creditworthiness.
5. Customer activity.

Removing redundant variables reduces computational cost and can improve model efficiency.

C. DATA TRANSFORMATION

i. Standardization

Standardization formula:

$$Z = (X - \mu) / \sigma$$

The attributes Cust_ID and Customer_Class are excluded.

The standardized values are approximately:

Customer	Age	Income	Purchase Value	Credit Score	Orders
Avg Balance	EMI				
U1	-1.50	-1.16	-1.12	-1.35	-1.06
-1.04					
U2	-0.53	-0.59	-0.55	-0.43	-0.58
-0.58					

U3	0.45	0.12	0.13	-0.06	0.34	-0.01
-0.12						
U4	1.24	1.12	1.37	1.42	1.21	1.20
1.28						
U5	-1.01	-1.01	-1.00	-0.79	-0.90	-0.90
-0.93						
U6	-0.04	-0.16	-0.21	-0.24	-0.16	-0.25
-0.23						
U7	1.63	1.97	1.82	1.79	1.83	2.01
1.97						
U8	-0.23	-0.30	-0.44	-0.33	-0.40	-0.41
-0.35						

ii. Need for Standardization

Standardization is required because the attributes have different numerical scales.

For example:

Age ranges from approximately 24 to 56.

Income ranges from ₹40K to ₹150K.

Credit Score ranges from 650 to 820.

EMI ranges from ₹5,000 to ₹18,000.

Without standardization, large numerical values may dominate machine-learning algorithms.

Therefore, standardization puts all numerical attributes on a comparable scale.

CONCLUSION

Standardization improves the performance of distance-based and optimization-based machine-learning algorithms.

Q3. DATA TRANSFORMATION AND DISCRETIZATION IN EDUCATION ANALYTICS (L3, L4, L5)

A. EQUAL-WIDTH BINNING

i. STUDY HOURS

Minimum = 2

Maximum = 11

Number of bins = 3

Formula:

Bin Width = (Maximum - Minimum) / Number of Bins

Bin Width = $(11 - 2) / 3$

Bin Width = $9 / 3$

Bin Width = 3

Therefore, the intervals are:

Bin 1 = 2–5

Bin 2 = 5–8

Bin 3 = 8–11

Assignment:

2 → Bin 1

3 → Bin 1

4 → Bin 1

5 → Bin 1

6 → Bin 2

7 → Bin 2

8 → Bin 2

9 → Bin 3

10 → Bin 3

11 → Bin 3

ii. TEST SCORE

Minimum = 40

Maximum = 100

Number of bins = 3

Formula:

Bin Width = $(100 - 40) / 3$

Bin Width = $60 / 3$

Bin Width = 20

Therefore, the intervals are:

Bin 1 = 40–60

Bin 2 = 60–80

Bin 3 = 80–100

Assignment:

40 → Bin 1

50 → Bin 1

55 → Bin 1

65 → Bin 2

75 → Bin 2

85 → Bin 3

90 → Bin 3

95 → Bin 3

98 → Bin 3

100 → Bin 3

B. EQUAL-FREQUENCY BINNING

There are 10 records.

For 3 bins, the records are divided approximately as:

$$4 + 3 + 3 = 10$$

STUDY HOURS

Bin 1:

2, 3, 4, 5

Bin 2:

6, 7, 8

Bin 3:

9, 10, 11

Therefore:

2 → Bin 1

3 → Bin 1

4 → Bin 1

5 → Bin 1

6 → Bin 2

7 → Bin 2

8 → Bin 2

9 → Bin 3

10 → Bin 3

11 → Bin 3

TEST SCORES

Bin 1:

40, 50, 55, 65

Bin 2:
75, 85, 90

Bin 3:
95, 98, 100

Therefore:

40 → Bin 1
50 → Bin 1
55 → Bin 1
65 → Bin 1
75 → Bin 2
85 → Bin 2
90 → Bin 2
95 → Bin 3
98 → Bin 3
100 → Bin 3

C. CONCEPT HIERARCHY CONSTRUCTION

STUDY HOURS

Study Hours → Low / Medium / High

Low:
2–5 hours

Medium:

6–8 hours

High:
9–11 hours

TEST SCORE

Test Score → Poor / Average / Excellent

Poor:
40–60

Average:
61–80

Excellent:
81–100

CONCLUSION

Discretization converts continuous numerical values into meaningful categories, making student-performance analysis easier to interpret.

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Q4. NORMALIZATION

Given data:

200, 300, 400, 600, 1000

(a) MIN-MAX NORMALIZATION

Formula:

$$X' = (X - X_{\min}) / (X_{\max} - X_{\min})$$

Minimum = 200

Maximum = 1000

Therefore:

$$X' = (X - 200) / (1000 - 200)$$

$$X' = (X - 200) / 800$$

For 200:

$$X' = (200 - 200) / 800$$

$$X' = 0$$

For 300:

$$X' = (300 - 200) / 800$$

$$X' = 100 / 800$$

$$X' = 0.125$$

For 400:

$$X' = (400 - 200) / 800$$

$$X' = 200 / 800$$

$$X' = 0.250$$

For 600:

$$X' = (600 - 200) / 800$$

$$X' = 400 / 800$$

$$X' = 0.500$$

For 1000:

$$X' = (1000 - 200) / 800$$

$$X' = 1.000$$

FINAL MIN-MAX RESULTS:

$$200 \rightarrow 0.000$$

$$300 \rightarrow 0.125$$

$$400 \rightarrow 0.250$$

$$600 \rightarrow 0.500$$

$$1000 \rightarrow 1.000$$

(b) Z-SCORE NORMALIZATION

Formula:

$$Z = (X - \mu) / \sigma$$

Step 1: Calculate Mean

$$\mu = (200 + 300 + 400 + 600 + 1000) / 5$$

$$\mu = 2500 / 5$$

$$\mu = 500$$

Step 2: Calculate Standard Deviation

$$\sigma = \sqrt{[(200-500)^2 + (300-500)^2 + (400-500)^2 + (600-500)^2 + (1000-500)^2] / 5}$$

$$\sigma = \sqrt{[90000 + 40000 + 10000 + 10000 + 250000] / 5}$$

$$\sigma = \sqrt{400000 / 5}$$

$$\sigma = \sqrt{80000}$$

$$\sigma = 282.843$$

Step 3: Calculate Z-Scores

For 200:

$$Z = (200 - 500) / 282.843$$

$$Z = -1.061$$

For 300:

$$Z = (300 - 500) / 282.843$$

$$Z = -0.707$$

For 400:

$$Z = (400 - 500) / 282.843$$

$$Z = -0.354$$

For 600:

$$Z = (600 - 500) / 282.843$$

$$Z = 0.354$$

For 1000:

$$Z = (1000 - 500) / 282.843$$

$$Z = 1.768$$

FINAL Z-SCORE RESULTS:

$$200 \rightarrow -1.061$$

$$300 \rightarrow -0.707$$

$$400 \rightarrow -0.354$$

$$600 \rightarrow 0.354$$

$$1000 \rightarrow 1.768$$

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Q5. DATA TRANSFORMATION IN ONLINE SHOPPING BEHAVIOR ANALYSIS (L3, L4, L5)

A. EQUAL-FREQUENCY BINNING

Given Visit Duration:

15, 30, 45, 60, 120, 300, 600, 900, 1800, 3600 seconds.

There are 10 records.

For 3 bins:

$$4 + 3 + 3 = 10$$

BIN 1 – SHORT VISITS

15, 30, 45, 60

BIN 2 – MEDIUM VISITS

120, 300, 600

BIN 3 – LONG VISITS

900, 1800, 3600

Assignment:

C1 → 15 seconds → Short

C2 → 30 seconds → Short

C3 → 45 seconds → Short

C4 → 60 seconds → Short

C5 → 120 seconds → Medium

C6 → 300 seconds → Medium

C7 → 600 seconds → Medium

C8 → 900 seconds → Long

C9 → 1800 seconds → Long

C10 → 3600 seconds → Long

ANALYSIS

Equal-frequency binning groups observations according to their rank rather than fixed numerical ranges.

It effectively separates:

Short visits: 15–60 seconds

Medium visits: 120–600 seconds

Long visits: 900–3600 seconds

Therefore, it is useful for customer segmentation.

B. Z-SCORE NORMALIZATION

Formula:

$$Z = (X - \mu) / \sigma$$

i. Calculate Mean

$$\mu = (15 + 30 + 45 + 60 + 120 + 300 + 600 + 900 + 1800 + 3600) / 10$$

$$\mu = 7470 / 10$$

$$\mu = 747 \text{ seconds}$$

Standard Deviation:

Using population standard deviation:

$$\sigma = \sqrt{[\Sigma(X - \mu)^2 / N]}$$

$$\sigma \approx 1090.92 \text{ seconds}$$

ii. Z-SCORE RESULTS

15 seconds	→ -0.671
30 seconds	→ -0.657
45 seconds	→ -0.643
60 seconds	→ -0.630
120 seconds	→ -0.575
300 seconds	→ -0.410
600 seconds	→ -0.135
900 seconds	→ 0.140
1800 seconds	→ 0.965
3600 seconds	→ 2.615

iii. Effect of Extreme Values

The values 1800 and 3600 seconds are much larger than most observations.

For 3600 seconds:

$$Z = 2.615$$

Therefore, it is clearly above the mean.

Z-score normalization does not remove the outlier. Instead, it represents the distance of the observation from the mean in terms of standard deviations.

C. MIN-MAX NORMALIZATION

Formula:

$$X' = (X - X_{\min}) / (X_{\max} - X_{\min})$$

$$\text{Minimum} = 15$$

$$\text{Maximum} = 3600$$

Therefore:

$$X' = (X - 15) / (3600 - 15)$$

$$X' = (X - 15) / 3585$$

Calculations:

For 15:

$$X' = (15 - 15) / 3585$$

$$X' = 0.000$$

For 30:

$$X' = (30 - 15) / 3585$$

$$X' = 0.004$$

For 45:

$$X' = (45 - 15) / 3585$$

$$X' = 0.008$$

For 60:

$$X' = (60 - 15) / 3585$$

$$X' = 0.013$$

For 120:

$$X' = (120 - 15) / 3585$$

$$X' = 0.029$$

For 300:

$$X' = (300 - 15) / 3585$$

$$X' = 0.079$$

For 600:

$$X' = (600 - 15) / 3585$$

$$X' = 0.163$$

For 900:

$$X' = (900 - 15) / 3585$$

$$X' = 0.247$$

For 1800:

$$X' = (1800 - 15) / 3585$$

$$X' = 0.498$$

For 3600:

$$X' = (3600 - 15) / 3585$$

$$X' = 1.000$$

FINAL MIN-MAX RESULTS:

15 seconds → 0.000

30 seconds → 0.004

45 seconds → 0.008

60 seconds → 0.013

120 seconds → 0.029
300 seconds → 0.079
600 seconds → 0.163
900 seconds → 0.247
1800 seconds → 0.498
3600 seconds → 1.000

COMPARISON OF Z-SCORE AND MIN-MAX NORMALIZATION

Z-SCORE NORMALIZATION

1. Does not have a fixed range.
2. Centers the data around mean = 0.
3. Standard deviation becomes 1.
4. Sensitive to extreme values.
5. Useful for statistical and machine-learning applications.

MIN-MAX NORMALIZATION

1. Converts values into the range 0 to 1.
2. Minimum value becomes 0.
3. Maximum value becomes 1.
4. Highly sensitive to extreme values.
5. Useful when a fixed numerical range is required.

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